

General- m Minimal-Equilibrium Stability

Proved stability regions, the bifurcation boundary, and an exact $m = 3$ counterexample

ShenWork formalization project

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Abstract

We study the one-dimensional Neumann minimal chemotaxis system with population mobility u^m , extending the $m = 1$ stability results in Chen–Ruau–Shen Part II. Three positive results are formalized: the full two-disjunct global-stability formula for $1 \leq m < 2$; its critical $m = 2$ version under an explicit endpoint admissibility condition; and a small-sensitivity theorem for every $m > 1$, obtained from good times for the negative-power energy and oscillation control of u^{1-m} .

The energy-method boundary $m = 2$ is not a boundary of the dynamics. A weakly nonlinear calculation gives the direction coefficient of the first Neumann steady branch. Its sign changes at $m_c(U) \approx 2.867, 2.937, 3.006$ for masses $U = 0.5, 1, 2$, respectively. A negative coefficient predicts a backward branch and hence a steady obstruction below the linear threshold. For the fixed tuple

$$m = 3, \quad \beta = \gamma = \mu = \nu = U = 1,$$

this obstruction is not merely numerical: an amplitude-factored implicit branch in a Wiener algebra, its negative direction coefficient, the positive classical steady state, and the failure of global attraction are all formalized in Lean 4.30.0 / Mathlib v4.30.0.

We distinguish rigorously between the proved $m = 3$ counterexample, the formalized sufficient stability regions, and the analytically/numerically located direction-change curve. In particular, the sign $m < m_c(U)$ by itself is not claimed to prove global stability up to the linear threshold.

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1 Model, mass constraint, and the stability question

On the unit interval $I = (0, 1)$, consider

$$u_t = u_{xx} - \chi_0 \partial_x \left(\frac{u^m}{(1+v)^\beta} v_x \right), \quad (1)$$

$$0 = v_{xx} - \mu v + \nu u^\gamma, \quad (2)$$

$$u_x(t, 0) = u_x(t, 1) = v_x(t, 0) = v_x(t, 1) = 0. \quad (3)$$

This is the minimal model $a = b = 0$. Integrating (1) and using (3) shows that

$$\int_0^1 u(t, x) dx = U$$

is conserved. On this mass hyperplane, the corresponding constant equilibrium is

$$u_* = U, \quad v_* = \frac{\nu}{\mu} U^\gamma. \quad (4)$$

The global-stability property formalized in the repository quantifies over positive global bounded classical solutions with this conserved mass. It has two conclusions:

1. uniform convergence $u(t) \rightarrow U$ in the supremum norm;
2. for each orbit, constants $C, \rho, t_0 > 0$ such that the population and signal converge exponentially in the stated C^1 -type norms for $t \geq t_0$.

The positive entry time is essential because the paper's data are not assumed C^1 at $t = 0$.

2 Linear threshold

Let $\lambda_n = n^2 \pi^2$, $n \geq 0$, be the Neumann eigenvalues. Linearize at (4), writing $u = U + p$, $v = v_* + q$. For mode $n \geq 1$, (2) gives

$$\hat{q}_n = \frac{\nu \gamma U^{\gamma-1}}{\mu + \lambda_n} \hat{p}_n.$$

The population growth rate is therefore

$$\sigma_n(\chi_0) = -\lambda_n + \chi_0 \frac{\nu \gamma U^{m+\gamma-1} \lambda_n}{(1+v_*)^\beta (\mu + \lambda_n)}. \quad (5)$$

For the unit interval, the threshold for the first nonzero mode is

$$\chi_{\text{lin}}(U, m) = \frac{(1+v_*)^\beta (\mu + \pi^2)}{\nu \gamma U^{m+\gamma-1}}. \quad (6)$$

Since $\mu + \lambda_n$ increases with n , the first mode is the first to lose stability. Thus $\chi_0 < \chi_{\text{lin}}$ is the relevant linearly stable half-line for the unit interval.

For the tuple used in the exact counterexample,

$$U = \beta = \gamma = \mu = \nu = 1,$$

so $v_* = 1$ and

$$\chi_{\text{lin}} = 2(1 + \pi^2) \approx 21.73920880. \quad (7)$$

3 Formalized positive stability results

3.1 The exact status table

Exponent range	Additional condition	Formalized conclusion
$1 \leq m < 2$	Either disjunct of the explicit full- m global-stability formula	Mass-constrained global sup convergence and orbitwise eventual exponential C^1 convergence
$m = 2$	The same formula plus $\chi_0 \sqrt{\mu} \sigma_\beta U \max\{2, \gamma\} < 1$	The critical endpoint version of the same conclusion
Every $m > 1$	Linear stability and the explicit small-sensitivity inequality associated with a certified local basin radius	The same global/eventual conclusion
$m = 3$, fixed tuple, some $0 < \chi_0 < \chi_{\text{lin}}$	None	Global attraction is false: an exact non-constant positive steady orbit exists

The first two rows are sufficient conditions, not the assertion that every linearly stable parameter is globally stable. The fourth row is an exact false-side theorem. Together they show that $m = 2$ is an analytic method boundary, not a universal dynamical threshold.

3.2 $1 \leq m < 2$: entropy and signal-energy disjuncts

The formal predicate `MinimalGlobalStabilityFormulaConditionFullM` packages two explicit conditions:

$$\text{minimal1 entropy condition} \quad \vee \quad \text{minimal2 signal-energy condition}.$$

The `minimal2` branch additionally has $\gamma = 1$. Both thresholds include the relevant linear-stability and nonlinear basin-entry inequalities, evaluated using the canonical eventual upper and signal-lower constants.

Theorem 1 (subcritical full formula). *Assume $1 \leq m < 2$, $a = b = 0$, $\beta \geq 1$, $U > 0$, and either full- m formula disjunct. Then $(U, \nu U^\gamma / \mu)$ is globally attracting among positive bounded mass- U orbits, with orbitwise eventual exponential C^1 convergence.*

This is `intervalDomainM_Theorem_2_5_EventualGlobalStabilityFormula` in `Paper3/IntervalDomainMMinimalSignalEnergyGlobal.lean`. Its two components are `intervalDomainM_Theorem_2_5_minimal1_EventualGlobalStabilityFormula` and `intervalDomainM_Theorem_2_5_minimal2_EventualGlobalStabilityFormula`.

3.3 The critical endpoint $m = 2$

At $m = 2$, the mass-seeded L^P recurrence becomes critical. The missing room is supplied by the explicit admissibility condition

$$\chi_0 \sqrt{\mu} \sigma_\beta U \max\{2, \gamma\} < 1, \quad \sigma_\beta = \begin{cases} 1, & \beta = 1, \\ \frac{(\beta - 1)^{\beta-1}}{\beta^\beta}, & \beta > 1. \end{cases} \quad (8)$$

Here σ_β is the sharp upper factor for $z/(1+z)^\beta$ on $z \geq 0$. Condition (8) leaves room for a restarted exponent strictly above two and at least γ ; the endpoint Agmon estimate then supplies the eventual box used by both formula branches.

Theorem 2 (critical full formula). *Assume $m = 2$, $a = b = 0$, $\beta \geq 1$, $U > 0$, (8), and either full- m formula disjunct. Then the same global and eventual exponential conclusion holds.*

The Lean capstone is `intervalDomainM_Theorem_2_5_critical_EventualGlobalStabilityFormula` in `Paper3/IntervalDomainMMinimalCriticalGlobal.lean`.

3.4 Every $m > 1$: the small-sensitivity route

This route is independent of the $m < 2$ interpolation budget. Its key quantity is the negative-power weighted gradient

$$G(t) = \int_0^1 u(t, x)^{-2m} u_x(t, x)^2 dx. \quad (9)$$

The entropy production provides arbitrarily late good times at which G is as close as desired to the explicit floor

$$\chi_0^2 \mu \sigma_\beta^2. \quad (10)$$

Since

$$\partial_x u^{1-m} = (1-m) u^{-m} u_x,$$

Cauchy–Schwarz on the unit interval gives the exact oscillation estimate

$$|u(t, x)^{1-m} - u(t, z)^{1-m}| \leq (m-1) \sqrt{G(t)}. \quad (11)$$

The mass constraint and continuity provide a point z with $u(t, z) = U$. Thus a good time controls u^{1-m} uniformly around U^{1-m} .

For a desired local supremum radius $0 < \delta < U$, define

$$r_m(U, \delta) = \min \left\{ U^{1-m} - (U + \delta)^{1-m}, \frac{(U - \delta)^{1-m} - U^{1-m}}{2} \right\}. \quad (12)$$

Because $1 - m < 0$, both entries are positive. If

$$(m-1) \chi_0 \sqrt{\mu} \sigma_\beta < r_m(U, \delta), \quad (13)$$

then a sufficiently late good slice satisfies

$$\|u(t, \cdot) - U\|_\infty < \delta.$$

Linear stability supplies a positive local basin radius δ ; after the good slice enters that basin, an autonomous time shift and the strong sectorial bootstrap give eventual exponential convergence.

Equivalently, the explicit sensitivity threshold is

$$\chi_0 < \frac{r_m(U, \delta)}{(m-1)\sqrt{\mu}\sigma_\beta}. \quad (14)$$

Theorem 3 (all- $m > 1$ small-sensitivity branch). *Assume $m > 1$, $a = b = 0$, $\beta \geq 1$, $U > 0$, linear stability, and (14) for a certified local basin radius $0 < \delta < U$. Then the constant equilibrium is globally attracting on the mass- U hyperplane, and every positive bounded orbit converges exponentially in C^1 after an orbit-dependent positive time.*

The principal Lean statements are `intervalDomainM_negativePower_oscillation_le`, `intervalDomainM_minimal_exists_late_supClose_smallSensitivity`, and `intervalDomainM_Theorem_2_5_supercritical_smallSensitivity`. The theorem `intervalDomainM_supercritical_smallSensitivity_exists_threshold` constructs an admissible δ from linear stability and exposes (14) as the remaining sufficient inequality.

4 The steady-state bifurcation mechanism

4.1 Integrated flux equation

To locate a false side, specialize first to

$$\beta = \gamma = \mu = \nu = 1.$$

At a positive steady state, the first equation and Neumann conditions give

$$u_x - \chi \frac{u^m}{1+v} v_x = 0.$$

Let $H'_m(u) = u^{-m}$:

$$H_m(u) = \begin{cases} \log u, & m = 1, \\ \frac{u^{1-m}}{1-m}, & m \neq 1. \end{cases}$$

Then every positive steady state satisfies

$$H_m(u) - \chi \log(1+v) = C, \quad -v_{xx} + v = u, \quad (15)$$

together with $\int_0^1 u = U$. This removes one derivative and is the equation used by both the weakly nonlinear calculation and the finite difference continuation.

4.2 Weakly nonlinear expansion

Let

$$e_1(x) = \cos(\pi x), \quad e_2(x) = \cos(2\pi x),$$

and define the elliptic response factors

$$r_1 = \frac{1}{1+\pi^2}, \quad r_2 = \frac{1}{1+4\pi^2}, \quad b = 1+U.$$

Seek a mass-preserving branch

$$u = U + \varepsilon e_1 + \varepsilon^2 a_2 e_2 + O(\varepsilon^3), \quad (16)$$

$$v = U + \varepsilon r_1 e_1 + \varepsilon^2 a_2 r_2 e_2 + O(\varepsilon^3), \quad (17)$$

$$\chi = \chi_{\text{lin}} + \varepsilon^2 \chi_2 + O(\varepsilon^3). \quad (18)$$

The first-order solvability condition gives

$$\chi_{\text{lin}} = \frac{b}{r_1 U^m}. \quad (19)$$

The second harmonic at order ε^2 gives

$$a_2 = \frac{m/U - r_1/b}{4(1 - r_2/r_1)}. \quad (20)$$

Projecting the third-order equation onto e_1 yields

$$\chi_2 = \chi_{\text{lin}} S(m, U), \quad (21)$$

$$S(m, U) = \frac{a_2}{2} \left(\frac{r_2}{b} - \frac{m}{U} \right) + \frac{m(m+1)}{8U^2} - \frac{r_1^2}{4b^2}. \quad (22)$$

At the level of this weakly nonlinear expansion:

$$S > 0 \implies \text{the local branch points to } \chi > \chi_{\text{lin}},$$

whereas

$$S < 0 \implies \text{the branch is backward and contains } \chi < \chi_{\text{lin}}.$$

If the formal branch is promoted to an actual nonconstant positive steady branch, every point on its backward side is a bounded global orbit that cannot converge to the constant equilibrium. The required analytic branch construction is supplied below for the fixed $m = 3, U = 1$ tuple; no all- m branch theorem is being asserted here.

4.3 Direction-change curve

Solving $S(m, U) = 0$ gives:

Mass U	$m_c(U)$	Sign for $m > m_c(U)$
0.5	2.866754572143	backward
1	2.936658202934	backward
2	3.005668668328	backward

These numbers are roots of the explicit coefficient (22), not fitted continuation parameters.

Remark 1 (scope of the curve). *The curve $S = 0$ is a local steady-bifurcation direction boundary. For $m > m_c(U)$, the coefficient predicts a direct obstruction below the linear threshold. The required local branch theorem and the resulting obstruction are carried out completely in Lean for the fixed $m = 3, U = 1$ tuple. For $m < m_c(U)$, a positive coefficient merely removes this predicted local obstruction. It does not prove global attraction for every $0 < \chi_0 < \chi_{\text{lin}}$.*

5 Numerical cross-check

The script `research_notes/m_supercritical_bifurcation_probe.py` performs two independent checks.

Discrete weak coefficient. For N finite-difference subintervals it replaces

$$\pi^2, 4\pi^2 \quad \text{by} \quad \lambda_{1,h} = \frac{4}{h^2} \sin^2 \frac{\pi}{2N}, \quad \lambda_{2,h} = \frac{4}{h^2} \sin^2 \frac{\pi}{N}$$

in (20)–(22).

Nonlinear solve. At every grid node it enforces the integrated flux equation (15), the finite-difference elliptic equation with reflected Neumann ghost points, the trapezoidal mass constraint, and a prescribed first cosine coefficient. It then compares the fitted quadratic branch direction with the discrete analytic coefficient.

For $N = 160$:

m	U	$\chi_{\text{lin},h}$	analytic $\chi_{2,h}$	fitted $\chi_{2,h}$	largest residual
4	1	21.7385746366	−4.24003585072	−4.24003584422	1.1×10^{-11}
3	0.5	130.431447819	−9.59085815306	−9.59085814853	4.8×10^{-12}

At relative amplitude 0.02, the sensitivity shifts are

$$\Delta\chi = -1.6966 \times 10^{-3} \quad (m = 4, U = 1),$$

and

$$\Delta\chi = -9.5917 \times 10^{-4} \quad (m = 3, U = 0.5).$$

Pseudo-arclength continuation preserves the backward direction through the reported steps. These computations validate the coefficient and the implementation, but numerical residuals are not used as proof of the exact $m = 3$ counterexample.

6 Exact Lean proof of the $m = 3$ false side

6.1 Functional setting

Fix

$$m = 3, \quad \beta = \gamma = \mu = \nu = U = 1.$$

The steady system is

$$0 = u_{xx} - \chi \partial_x \left(\frac{u^3 v_x}{1 + v} \right), \tag{23}$$

$$0 = v_{xx} - v + u. \tag{24}$$

The formalization works in a weighted ℓ^1 Wiener algebra of Fourier coefficients. This choice has four useful properties:

1. multiplication is continuous, so nonlinear products remain in the algebra;

2. the coefficient norm controls pointwise evaluation;
3. two Wiener weights control the two spatial derivatives needed for a classical steady state;
4. even real sequences encode Neumann cosine profiles, and the mean-zero subspace enforces the mass constraint.

Split the mean-zero even profile into its first cosine mode and a closed complement:

$$h = e_1 + z, \quad z \perp e_1.$$

With amplitude a ,

$$u_a = 1 + ah, \quad v_a = 1 + aRh,$$

where $R = (-\partial_{xx} + 1)^{-1}$ is the diagonal Fourier resolver. Dividing the steady residual by the amplitude produces a smooth blown-up map in (a, z, χ) . At

$$(a, z, \chi) = (0, 0, 2(1 + \pi^2))$$

its derivative in (z, χ) is a bounded linear isomorphism. The Banach implicit function theorem therefore yields a C^3 branch

$$a \longmapsto (z(a), \chi(a)).$$

The construction and inverse are in `MinimalSteadyWABlowup.lean`, `MinimalSteadyWALinearIso.lean`, and `MinimalSteadyWAImplicitBranch.lean`.

6.2 Exact branch direction

After multiplying out the harmless signal denominator, the branch satisfies the polynomial flux identity

$$(2 + aRh) Dh = \chi(a)(1 + ah)^3 DRh. \quad (25)$$

The mode-two component of the first derivative determines the second-harmonic correction. The mode-one component of the second derivative gives

$$\chi'(0) = 0, \quad \chi''(0) = -\frac{6\pi^4 + 37\pi^2 + 25}{24\pi^2(\pi^2 + 1)} < 0. \quad (26)$$

Since the expansion convention (18) has $\chi_2 = \chi''(0)/2$, (26) equals

$$\chi_2 \approx -0.189271408,$$

exactly matching (22) at $m = 3, U = 1$.

The formal theorems are `minimalSensitivityFirstJet_eq_zero`, `minimalSensitivitySecondJet_eq`, and `minimalSensitivitySecondJet_neg` in `Paper3/MinimalSteadyWADirection.lean`. Elementary derivative tests then show that $\chi(a)$ is strictly decreasing for all sufficiently small $a > 0$, while continuity keeps it positive:

$$0 < \chi(a) < 2(1 + \pi^2). \quad (27)$$

This selection is formalized in `Paper3/MinimalSteadyWABranchSelection.lean`.

6.3 From the Banach branch to a classical counterexample

For sufficiently small positive a , the Wiener norm bound gives

$$u_a(x) > 0, \quad v_a(x) > 0 \quad \text{for all } x.$$

Coefficient evaluation and Fourier differentiation turn the algebraic residual into the pointwise classical equations (23)–(24). Evenness gives the Neumann boundary conditions, the missing zero Fourier coefficient gives

$$\int_0^1 u_a(x) dx = 1,$$

and the fixed first coefficient makes u_a nonconstant.

Theorem 4 (exact subcritical steady state). *There exist $0 < \chi < 2(1 + \pi^2)$ and positive, nonconstant, C^2 Neumann profiles u, v satisfying (23)–(24) and $\int_0^1 u = 1$.*

This is `exists_nonconstant_minimal_steady_below_threshold_m3` in `Paper3/MinimalSteadyWACounterexample.lean`.

View the profiles as constant-in-time functions. They form a positive, bounded, mass-one global classical orbit. If the constant equilibrium $(1, 1)$ attracted every such orbit, then $u(x)$ would converge both to its constant-in-time value $u(x)$ and to 1. Uniqueness of limits would force $u \equiv 1$, contradicting nonconstancy.

Theorem 5 (global attraction is false below the linear threshold). *For the fixed $m = 3$ tuple, there exists $0 < \chi < \chi_{\text{lin}}$ such that the minimal equilibrium does not attract all positive bounded mass-one classical orbits.*

The exact Lean theorem is `minimal_equilibrium_global_stability_false_m3`. This completes the false side inside the proof assistant; no unformalized Crandall–Rabinowitz theorem or numerical existence assumption remains in this tuple.

7 What is resolved, and what is not claimed

The results establish the following.

1. The $m < 2$ energy method extends the paper’s $m = 1$ result to $1 \leq m < 2$ under the full explicit formula conditions.
2. The critical exponent $m = 2$ is included under the explicit endpoint admissibility (8).
3. For every $m > 1$, sufficiently small sensitivity in the explicit sense (14) gives global attraction, provided the equilibrium is linearly stable.
4. The assertion “every linearly stable minimal equilibrium is globally attracting for every $m > 1$ ” is false. It already fails at $m = 3$ for a sensitivity strictly below χ_{lin} .
5. The direction curve $m_c(U)$ explains why the false side begins near $m \approx 2.9$ in the normalized family and why the energy boundary $m = 2$ is not sharp dynamically.

What is *not* proved is a complete necessary-and-sufficient global-stability diagram in the region between the sufficient thresholds and the subcritical steady branch. In particular:

- $m < m_c(U)$ does not imply global stability all the way to χ_{lin} ;
- $m > m_c(U)$ makes the weak coefficient negative; the corresponding local steady obstruction is proved here only for the fixed $m = 3, U = 1$ tuple and does not classify every smaller sensitivity;
- the three numerical values of $m_c(U)$ belong to the normalized $\beta = \gamma = \mu = \nu = 1$ family, not to arbitrary parameter tuples.

These qualifications are part of the result: they separate exact theorems from the phase diagram suggested by bifurcation analysis.

8 Lean theorem map

Component	Principal source / theorem
$1 \leq m < 2$, full formula	<code>IntervalDomainMMinimalSignalEnergyGlobal.lean:</code> <code>intervalDomainM_Theorem_2_5_EventualGlobalStabilityFormula</code>
$m = 2$, critical endpoint	<code>IntervalDomainMMinimalCriticalGlobal.lean:</code> <code>intervalDomainM_Theorem_2_5_critical_EventualGlobalStabilityFormula</code>
All $m > 1$, negative-power good slice	<code>IntervalDomainMNegativePowerBox.lean:</code> <code>intervalDomainM_minimal_exists_late_supClose_smallSensitivity</code>
All $m > 1$, small-sensitivity capstone	<code>IntervalDomainMSupercriticalSmallSensitivity.lean:</code> <code>intervalDomainM_Theorem_2_5_supercritical_smallSensitivity</code>
$m = 3$ Banach branch and direction	<code>MinimalSteadyWAImplicitBranch.lean,</code> <code>MinimalSteadyWADirection.lean,</code> <code>MinimalSteadyWABranchSelection.lean</code>
Exact classical steady state	<code>MinimalSteadyWACounterexample.lean:</code> <code>exists_nonconstant_minimal_steady_below_threshold_m3</code>
Failure of global attraction	<code>MinimalSteadyWACounterexample.lean:</code> <code>minimal_equilibrium_global_stability_false_m3</code>
Analytic/numerical cross-check	<code>research_notes/m_supercritical_bifurcation_probe.py</code>

All Lean capstones above are `sorry`-free. Their printed axiom audits use only the standard classical axioms `propext`, `Classical.choice`, and `Quot.sound`. The numerical script is reproducible but is supplementary; the $m = 3$ false-side theorem does not depend on it.